

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 3B

Understanding variation



Name _____ Class _____ Date _____

CLASS NOTES

Calculator allowed. Show your working and include units. Keep unrounded values in your calculations; round final SDs and CVs to **2 decimal places**. Task numbers match the PPT.

REVISION

One complete shift

Slides 3-4

These are the waits of **all five customers** in Shift A: **6, 10, 10, 10, 14 minutes**.

1. Calculate the mean and range.

MEAN: TOTAL $\div$ COUNT

RANGE: LARGEST – SMALLEST

2. For this particular shift, are these data a population or a sample? Explain.

GROUP AND REASON

EXERCISE B1a

Guided practice: deviations

Slides 8-10

A **deviation** is one observation minus the mean. Negative means below the mean; positive means above it. A deviation of zero means at the mean.

3. The mean is **10 minutes**. Finish each subtraction and total the deviations.

Wait (min)	Calculation: wait – mean	Deviation (min)
6	6 – 10	–4
10	_____	_____
10	_____	_____
10	_____	_____
14	_____	_____
Total	_____	_____

4. Does a total deviation of zero mean everyone waited the same time? Explain.

EXPLANATION

FROM DEVIATIONS TO VARIANCE AND STANDARD DEVIATION

EXERCISE B1a

Guided practice: square the deviations

Slides 12-13

A **squared deviation** is a deviation multiplied by itself. For example, $(-4)^2 = 16$.

5. Complete the final column and add its entries. Use the same mean of 10 minutes.

Wait (min)	Deviation (min)	Squared deviation (min ²)
6	-4	16
10	0	_____
10	0	_____
10	0	_____
14	+4	_____
Total	0	_____

WORKED EXAMPLE

Where does the mean get subtracted?

Slides 14-15

Population variance is the average of the squared deviations. Here, μ (mu) = 10 and $N = 5$. Subtract 10 from **each observation** before squaring:

$$\sigma^2 = \frac{(6 - 10)^2 + (10 - 10)^2 + (10 - 10)^2 + (10 - 10)^2 + (14 - 10)^2}{5}$$

$$\sigma^2 = \frac{16 + 0 + 0 + 0 + 16}{5} = \frac{32}{5} = 6.4 \text{ min}^2$$

The total **32** already contains all five subtractions and squares. Do not subtract the mean again from 32. Σ (capital sigma) means **add all the terms**.

Standard deviation (SD) is the square root of variance. It returns the result to the original units:

$$\sigma = \sqrt{\sigma^2} = \sqrt{6.4} \approx 2.53 \text{ min}$$

What changes?	Deviation	Standard deviation
Describes	One observation	Spread across the dataset
Possible sign	Negative, zero or positive	Zero or positive
Here (minutes)	$6 - 10 = -4$	About 2.53

The **mean** tells us the average wait. The **SD** tells us how spread out the waits are.

POPULATION: CALCULATE AND INTERPRET**EXERCISE B1b****Independent practice: another shift**

Slides 18-19

These are the waiting times of **all eight customers** in another complete shift.

7, 12, 9, 14, 8, 11, 15, 10 minutes

1. Calculate the population mean. Use this exact value throughout the table.

MEAN: SHOW THE TOTAL AND DIVISOR

2. Complete the working table. Subtract your mean, then square each deviation.

Wait (min)	Deviation: wait – mean (min)	Squared deviation (min ²)
7		
12		
9		
14		
8		
11		
15		
10		
Total		

3. Calculate the population variance and standard deviation. Show the divisor.

POPULATION VARIANCE (min²)

POPULATION SD (min): ROUND TO 2 DP

4. Compare this shift with **Shift A: mean 10 min, SD 2.53 min**. Which has the shorter average wait? Which has the more consistent waits?

COMPARE BOTH THE AVERAGE AND THE SPREAD

POPULATION AND SAMPLE: CHOOSE THE RIGHT METHOD

REFERENCE

Same process, different purpose

Slides 21-25

A **population** is the complete target group. A **sample** contains selected observations from it. An **estimate** is a calculated approximation of an unknown population value.

Measure	Population: the whole group	Sample: estimate the larger group
Count	N: capital N, population count	n: lowercase n, sample count
Mean	μ : mu, population mean $\mu = \frac{\sum x_i}{N}$	$\bar{x}$: x-bar, sample mean $\bar{x} = \frac{\sum x_i}{n}$
Variance	σ^2 : sigma squared $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$	s^2 : s squared $s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$
SD	σ : sigma, population SD $\sigma = \sqrt{\sigma^2}$	s : sample SD $s = \sqrt{s^2}$

x_i (x sub i) = one observation. Σ (capital sigma) = add the terms.

σ (lowercase sigma) is a different symbol: population standard deviation.

WHY $n - 1$?

The sample mean is estimated from the same data. Dividing by n tends to underestimate population variance; $n - 1$ adjusts for this. **Both means still divide by the count.**

EXERCISE B2a

Guided practice: estimate the month

Slides 26-27

The target group is **all 2,000 customers in a month**. Eight randomly selected customers give the following results. The mean and squared-deviation total are supplied.

$$n = 8, \quad \bar{x} = 15 \text{ min}, \quad \sum (x_i - \bar{x})^2 = 112 \text{ min}^2$$

1. Identify the target population and explain which variance divisor to use.

POPULATION AND DIVISOR

2. Calculate the sample variance, then the sample standard deviation.

SAMPLE VARIANCE (min²)

SAMPLE SD (min)

3. What do the mean and SD estimate about the month?

AVERAGE AND SPREAD

SAMPLE: CALCULATE AND INTERPRET

EXERCISE B2b

Independent practice: a new sample

Slides 28-29

A new **random sample of eight customers** is used to estimate the month's waiting times.

25, 18, 31, 22, 28, 21, 30, 26 minutes

1. Choose the method and explain why. **2.** Find the mean and complete the table.

POPULATION OR SAMPLE METHOD? WHY? 	MEAN: TOTAL ÷ COUNT
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Keep the **exact mean** and squared deviations. Round only the final variance and SD.

Wait (min)	Deviation: wait – mean (min)	Squared deviation (min ²)
25		
18		
31		
22		
28		
21		
30		
26		
Total		

3. Calculate the sample variance and SD. Give both to 2 decimal places.

SAMPLE VARIANCE (min²): SHOW THE DIVISOR 	SAMPLE SD (min): TAKE THE SQUARE ROOT
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4. Interpret both results: what do the sample mean and SD suggest about the month's waiting times?

AVERAGE WAIT AND SPREAD AROUND IT

COEFFICIENT OF VARIATION: COMPARE RELATIVE SPREAD

REFERENCE

Coefficient of variation (CV)

Slides 31-32

CV expresses the standard deviation as a percentage of the mean. **Absolute variation** uses the original units; **relative variation** compares spread with the mean.

$$CV_{\text{population}} = \frac{\sigma}{\mu} \times 100\% \quad CV_{\text{sample}} = \frac{s}{\bar{x}} \times 100\%$$

Use CV for positive measurements with a meaningful zero and a mean away from zero. Here, **0 mL means no liquid**.

Worked example: mean 50 mL, SD 5 mL gives CV = 10%.

Your turn: mean 100 mL, SD 5 mL. CV = _____%.

EXERCISE B3a

Guided practice: relative spread

Slides 33-34

A sample of bottle fills has **mean 137.5 mL** and **SD 4.4 mL**. Put the correct values into the formula and calculate CV.

$$CV = \frac{?}{?} \times 100\%$$

SUBSTITUTION AND CALCULATION

The standard deviation is _____% of the mean.

CV describes **relative spread**; it is not the percentage of defective bottles.

EXERCISE B3b

Independent practice: compare fills

Slides 35-36

Bottle type	Sample mean (mL)	Sample SD (mL)
Small	248.6	6.8
Large	742.3	18.9

1. Which bottle type has the lower absolute SD? _____

2. Calculate each CV. Give your final percentages to 2 decimal places.

SMALL BOTTLES: CV CALCULATION

LARGE BOTTLES: CV CALCULATION

3. Which has less relative variation? Explain using your CV results.

RELATIVE COMPARISON

KURTOSIS: EXTREME VALUES AND TAIL PATTERNS**REFERENCE****Read the shape as well as the SD**

Slides 38-39

Tails are the outer regions containing unusually low or high values. The **normal distribution** is a symmetric, bell-shaped reference. **Skewness** describes left-right asymmetry.

Kurtosis describes the influence of extreme values relative to spread. We use **ordinary kurtosis**: the normal reference has value **3**.

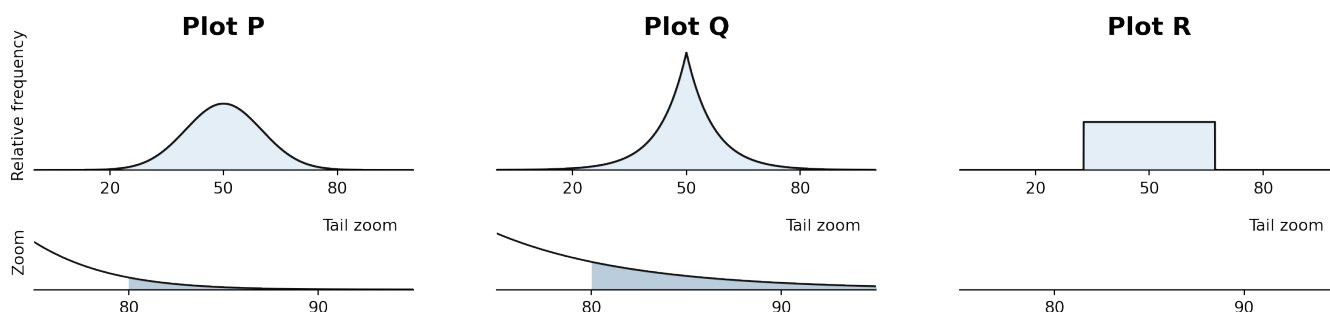
Term	Ordinary kurtosis	Compared with normal
Mesokurtic	3	Same kurtosis benchmark
Leptokurtic	Above 3	Greater influence of extremes
Platykurtic	Below 3	Lower influence of extremes

Peak height alone does not determine kurtosis. Here we interpret supplied values; no kurtosis calculation is required.

EXERCISE B4**Interpret the tail patterns**

Slides 40-41

These symmetric distributions have the **same mean and standard deviation**. Look at the full curves and the enlarged far-tail views.



1. Which plot has the greatest influence from extreme values? Use the tails to explain.

PLOT AND REASON

2. Correct this claim: "Equal SDs mean the distributions have the same shape."

CORRECTED STATEMENT

During feedback: record the supplied ordinary kurtosis values and the matching terms.

P: kurtosis and term	Q: kurtosis and term	R: kurtosis and term

EXIT CHECK AND OPTIONAL CHALLENGE

EXERCISE B5

Independent exit check

Slides 42-43

A new **random sample of ten customers** gives these summary results:

$$n = 10, \quad \bar{x} = 24 \text{ min}, \quad \sum (x_i - \bar{x})^2 = 90 \text{ min}^2$$

1. Calculate variance and SD. Explain your choice of divisor.

SAMPLE VARIANCE (min²)

SAMPLE SD (min): ROUND TO 2 DP

WHY THIS DIVISOR?

2. Find the deviation of a **19-minute wait**. Explain its sign.

SUBTRACTION AND MEANING

3. Correct this claim: “Every observation is exactly one standard deviation from the mean.”

CORRECTED STATEMENT

EXERCISE B6

Optional challenge: work backwards

Slides 44-45

Six values have **mean 7** and **sample standard deviation 2**.

6, 7, 7, 8, ?, ?

Find the two missing values. Use **both** the mean and the standard deviation.

Hint: use the mean to find the total of all six values. Use sample SD to find the total of their squared deviations. Show how both conditions are satisfied.

WORKING AND CHECK
